Cross-sectional intratumoral microbial abundance cannot establish causation

A calibration study using an established carcinogen as a natural control

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1 The question the retraction did not answer

The 2024 retraction of a pan-cancer tumor microbiome analysis established that reported signal can be human read misclassification and batch effect. The field responded with decontamination pipelines and curated resources. Those address one question: is a detected taxon really present. They do not address a second: if it is present, is it causal, and in which direction does the association run. The dominant design, cross-sectional tumor versus adjacent normal, detects a difference in abundance well and interprets it poorly, because it fixes neither the cause nor the temporal order.

2 Standard tests pass on data with no biology

10 of 10 seeds in which a cohort built to contain no biological signal passed both standard tests

Comparison against a no-information rate and label permutation both passed on zero-signal data. Only a confounder baseline and within-batch cross-validation discriminated. Batch-outcome confounding alone drove accuracy to 2.5 times chance on data with no biology.

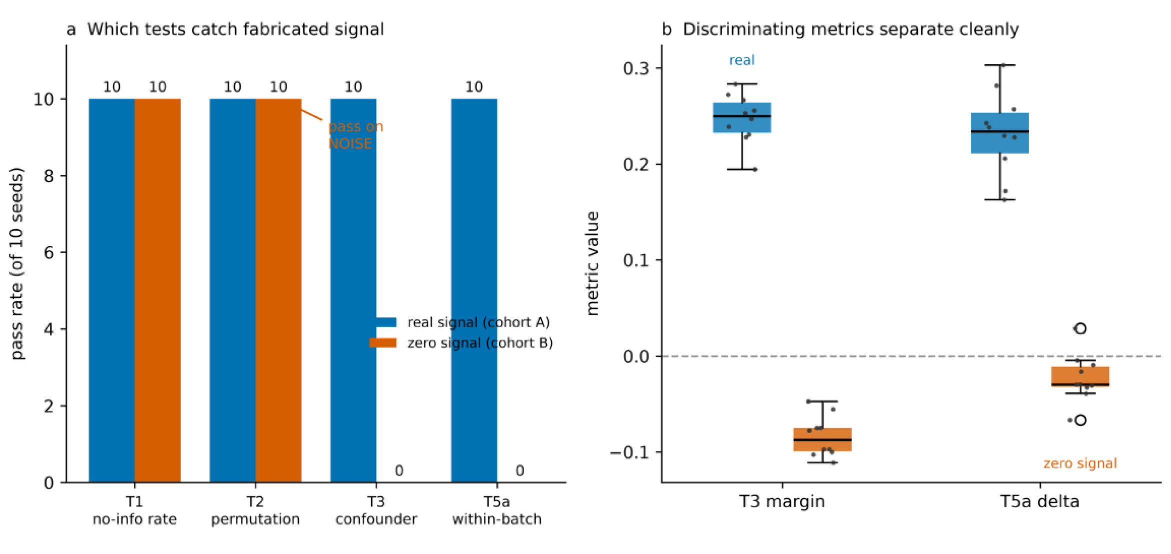

Fig 1 Pass rate by test across ten seeds. T1 and T2 pass in both cohorts; only T3 and T5a separate them.

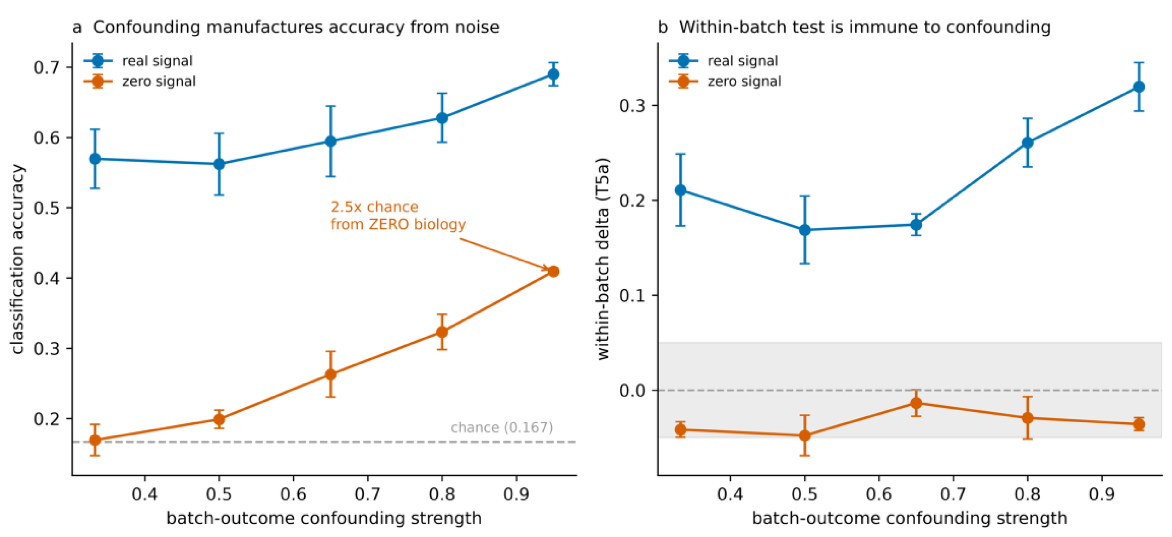

Fig 2 Accuracy against confounding strength. Within-batch delta stays near zero for zero-signal data at every level.

3 An established carcinogen runs backwards

8x depletion of H. pylori at the gastric tumor site
39 matched pairs, diff -0.99 log units, p 4.5e-4

Helicobacter pylori is an IARC Group 1 carcinogen and the accepted cause of gastric adenocarcinoma. In a decontaminated reference cohort it is depleted, not enriched, at the tumor site. The mechanism is general: gastric carcinogenesis proceeds through atrophy and intestinal metaplasia, which eliminate the acid-adapted mucosa the organism requires. It causes the disease and is then displaced by it. For the one organism whose causal role is known, cross-sectional abundance gives the wrong direction.

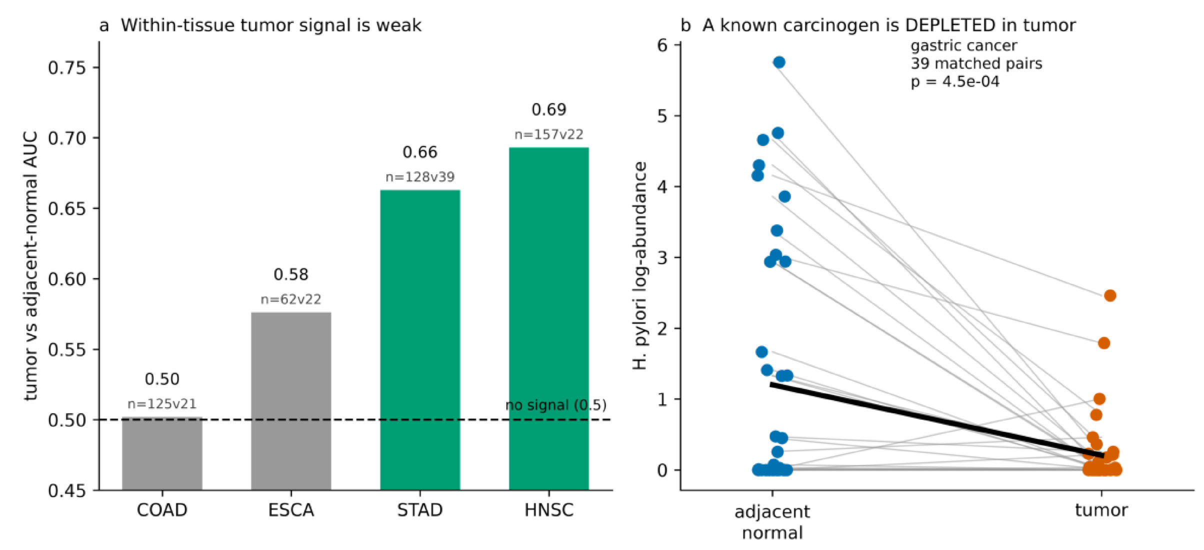

Fig 4 (a) Within-tissue tumor versus adjacent-normal discrimination, weak in all four evaluable cancers. (b) Paired H. pylori depletion.

4 Two ways clean data still misleads

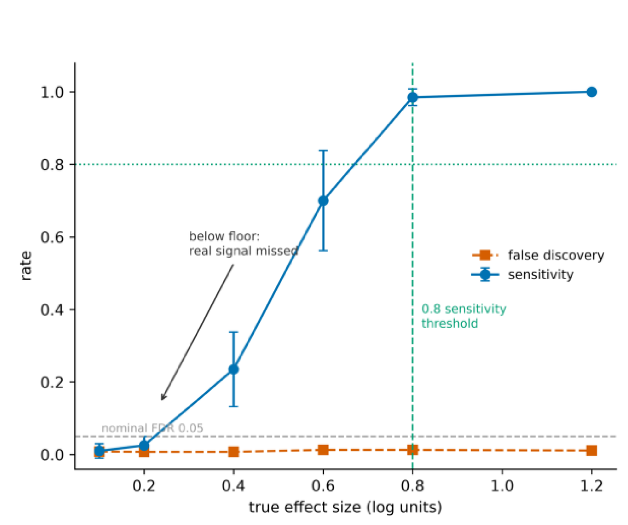

Fig 3 Detection floor is about 0.8 log units. A null must be reported as no signal above the floor.

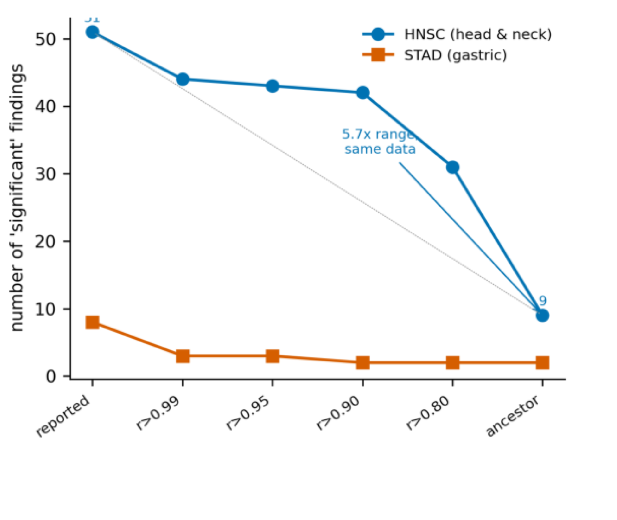

Fig 5 Discovery counts vary up to 5.7-fold with the taxonomic redundancy rule, on identical data.

DATA AND METHODS

Simulation Two synthetic cohorts, six cancer types, three batches, 150 taxa, 60 samples per type. One carries genuine signal, one carries none. Batch confounded with outcome at 0.75.
Audit T1 no-information rate, T2 label permutation, T3 confounder baseline, T5a within-batch cross-validation. Random forests.
Real data The Cancer Microbiome Atlas: 611 samples, 14,492 taxa, five TCGA projects, three centers.
MR MiBioGen, 211 taxa, 18,340 individuals. IVW primary, plus MR-Egger, weighted median, Cochran Q, BH-FDR.

5 Germline anchoring: the causal screen

100,204 case independent replication cohort
211 gut taxa, two-sample MR

Every FinnGen lead failed replication. The one FDR-significant hit, phylum Cyanobacteria, was diagnosed as pleiotropy via the NOS2 instrument rs2314810 and flipped direction on replication. A Bifidobacterium signal collapsed on removal of rs182549 at LCT. Zero taxa survive FDR in the large cohort.

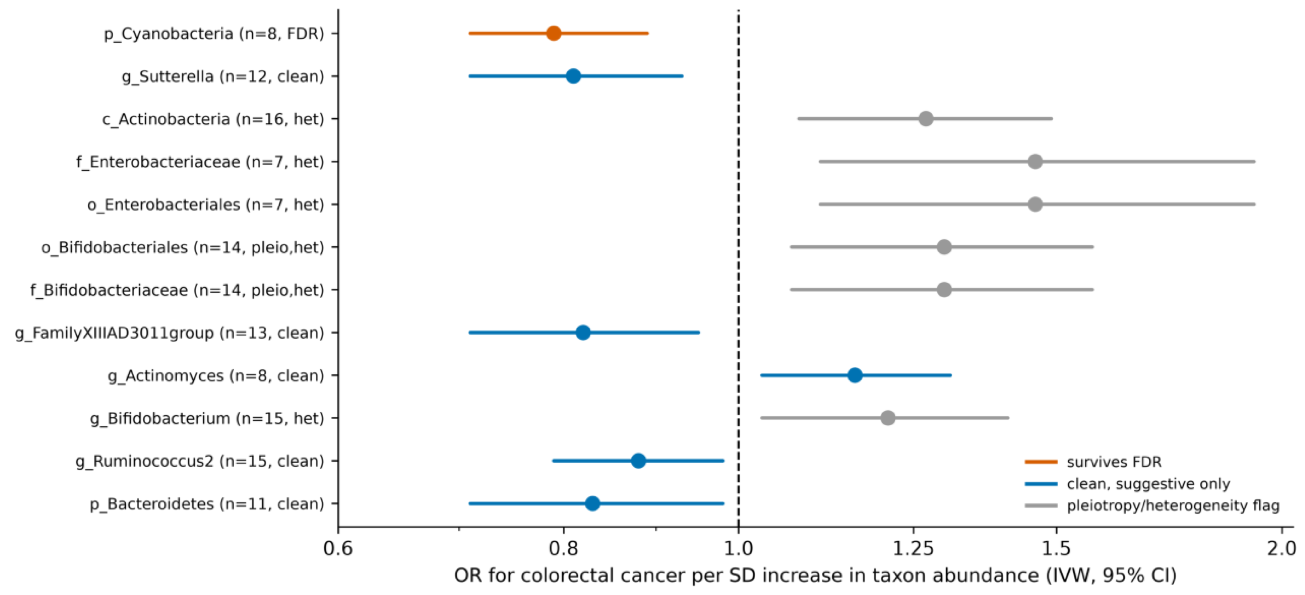

Fig 6 Two-sample MR of 211 gut taxa against colorectal cancer, FinnGen R12.

6 The signal is real but diffuse

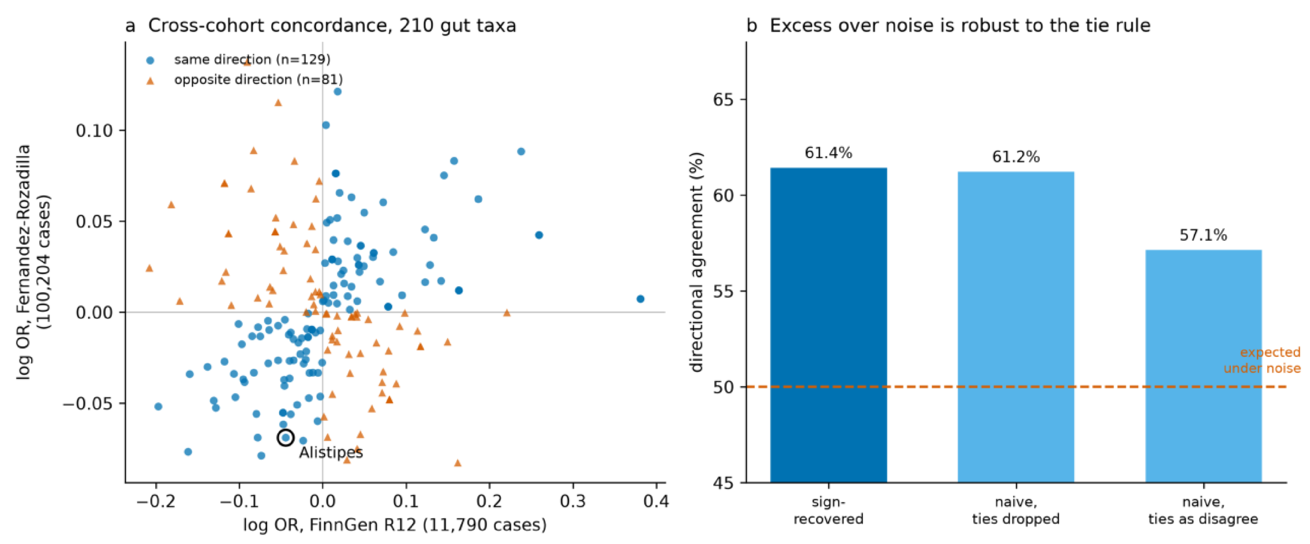

Fig 7 Cross-cohort concordance, 210 taxa, sign recovered. Agreement 61.4 percent (129/210, binomial p 0.0011) against 50 percent under noise, robust to the tie rule.

Alistipes points protective in three of three tests, nominal in two of three. Colorectal FinnGen is null at p 0.578. No test survives FDR. A lead requiring species-level instrumentation, not a finding.

CONCLUSION

Decontamination is necessary and not sufficient. Cross-sectional abundance cannot establish causation and, on the one organism whose causal role is known, gets the direction backwards. Causal inference here requires germline anchoring or temporal ordering, and both require an audit first.

Scan for the repository:
code, results, figures,
and the corrections log.

Methods Two synthetic cohorts, six cancer types, three batches, 150 taxa, 60 samples per type; one with genuine signal, one without, batch confounded with outcome at 0.75. Audit: T1 no-information rate, T2 label permutation, T3 confounder baseline, T5a within-batch CV.
Data The Cancer Microbiome Atlas, DOI 10.7924/r4bk1j35s, 611 samples, 14,492 taxa | MiBioGen, 211 taxa | FinnGen R12 | Fernandez-Rozadilla 2023, GCST90129505
Code Versioned pipeline, ten-seed determinism, input checksums, assertion-checked outputs, re-derived by CI on every push. github.com/julian-borges-md/tumor-microbiome-causality-audit

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